

Day 6: Introduction to Network Devices Theory (1 hour):Overview of switches, routers, hubs, firewalls, and access points. **Practical (1 hour):** Identify ports and functions of network devices in a setup. **Task:** Create a table comparing network devices and their features.

Task 1.

Create a table listing the main functions and typical ports used by switches, routers, hubs, firewalls, and access points in a home WiFi setup.

Device	Main Function	Typical Ports / Interfaces
Switch	Connects multiple wired devices in the same LAN and forwards data to the correct device.	Ethernet ports (RJ-45), Console port, Power
Router	Connects the local network to the Internet and routes packets between different networks.	WAN Ethernet, LAN Ethernet, Console, Power
Hub	Connects multiple devices and broadcasts incoming data to all connected ports.	Ethernet ports (RJ-45), Power
Firewall	Monitors and filters network traffic and blocks unauthorized connections.	WAN/LAN Ethernet, Console/Management, Power
Access Point (AP)	Provides wireless Wi-Fi connectivity and connects wireless devices to the wired network.	Ethernet, Power, Antenna(s)

Task 2.

Given images or diagrams of network devices, label each device and mark the ports (Ethernet, console, power, antenna) for at least three devices.



Device 1: Wireless Wi-Fi Router

WAN Port (Blue RJ-45): Connects to the ISP cable or modem to receive external internet.

LAN Ports (Yellow RJ-45, 1–4): Connect local wired devices like desktop PCs or gaming consoles.

Power Port (DC Jack): Plugged into the wall adapter for power supply.

Antennas (Left & Right): Broadcast and receive wireless Wi-Fi signals.

Device 2: Enterprise Network Switch

- **Ethernet Ports (RJ-45, 1–6):** Connect wired workstations, servers, and secondary access points using MAC addressing.
- **Console Port / USB:** Used by administrators for local command-line configuration and management.
- **Power Connector:** Provides AC power to the unit.

Device 3: Ethernet Desktop Switch

- **Ethernet Ports (RJ-45, 1–8):** Used to attach network patch cables from client machines to extend local LAN connectivity.

- **Power Port (Side/Rear):** Accepts DC power input.

Task 3.

Pick a popular app you use daily (like Instagram, Zomato, or Paytm) and describe in 3-4 lines how at least two different network devices (from switches, routers, firewalls, access points) play a role when you use that app.

- When you use Instagram on your phone, the phone first connects to the Wi-Fi access point, which provides wireless connectivity.
- The router receives the data and routes it from your home network to the Internet and Instagram's servers.
- A firewall can inspect the traffic and block unauthorized connections. In larger networks, a switch connects wired devices such as the router, access point, and computers.

Task 4.

Suppose your college WiFi is slow and sometimes disconnects. Based on the features of access points, switches, and routers, suggest one possible reason for the issue and how upgrading one device could help.

Hint: Think about device capacity, coverage, and traffic handling.

Possible reason: The college may have too few access points, causing too many students to connect to the same AP. The AP can become overloaded, resulting in slow speeds and frequent disconnections.

Possible solution: Upgrade or add higher-capacity Wi-Fi access points and distribute them properly across classrooms and common areas. This increases the number of devices that can be handled and improves Wi-Fi coverage. A higher-capacity switch/router may also be required if the wired network is becoming a traffic bottleneck.